



The impact of a supplementary preschool mathematics curriculum on children's early mathematics learning[☆]

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ABSTRACT

Scientifically-validated curricula, which demonstrate positive child outcomes, are the heart of high-quality preschools. The present study examines the impact of supplementing the whole-child comprehensive HighScope curriculum with the *Numbers Plus Preschool Mathematics* curriculum, training, and coaching (hereafter, *Numbers Plus*), using a multi-site cluster randomized trial (MSCRT) consisting of four levels: children, classrooms, sites, and states. The study is the first effort to examine the effect of supplementing a widely used comprehensive whole-child curriculum with a newly developed skill-based curriculum, where both were created by the same developer. Twenty-two sites, 52 classrooms, and 583 children (323 treatment; 260 control) in three states (Michigan, Maine, and Ohio) were selected to participate in the study. The findings show promise in *Numbers Plus* as an evidence-based, skill-based supplement to the whole-child comprehensive HighScope curriculum. The frequency of *Numbers Plus* small-group times, topics covered, and coaching contacts reached the intended dosage, indicating feasibility of implementation. In addition, while all children increased their mathematics knowledge and skills during their preschool year, we found an additional positive effect of *Numbers Plus* on children's math outcomes. However, the effect size was small. The study provides future directions for developers of a comprehensive whole-child curricula like HighScope to ensure evidence-based curriculum improvement and development.

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1. Introduction

A high-quality early childhood education is one of the most examined and proven educational interventions for children at risk of school failure. From small rigorous efficacy studies confined to controlled samples receiving intense services such as the Perry Preschool study and Abecedarian study (e.g., Campbell & Ramsey, 1994; Schweinhart et al., 2005) to scaled-up districtwide and statewide implementations such as the Chicago Parent-Child Centers, Boston Public Schools Universal Prekindergarten, Tulsa Universal Prekindergarten, New Jersey Abbott Preschool Program, North Carolina Smart Start and More at Four programs, and Michigan Great Start Readiness Program (e.g. Dodge, Bai, Ladd, & Muschkin, 2017; Frede, Jung, Barnett, Lamy, & Figueras, 2007; Frede, Jung, Barnett, & Figueras, 2009; Gilliam & Zigler, 2004; Gormley, Gayer, Phillips, & Dawson, 2005; Gormley, Phillips, & Gayer, 2008; Ladd, Muschkin, & Dodge, 2014; Reynolds & Temple, 1998; Reynolds, Temple, White, Ou, & Robertson, 2011; Schweinhart, Xiang, Daniel-Echols, Browning, & Wakabayashi, 2012; Weiland & Yoshikawa, 2013), the benefits of high-quality

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early childhood education is backed by five decades of research. Researchers and policymakers now agree that the early years are crucial to children's development, and high-quality preschools help set the stage for children's positive trajectories in school and in life.

Scientifically-validated curricula, which demonstrate positive child outcomes, are the heart of high-quality preschools. Hence, as new curricula are being developed and introduced to the field, more scrutiny is placed on their evidence. Head Start's Preschool Curriculum Consumer Report (National Center on Quality Teaching and Learning, 2015), for example, rates curricula on thirteen "components of effective curricula," which include curriculum effects on child outcomes, comprehensiveness and depth in learning domains, individualized instruction, ongoing assessment, and professional learning (NCQTL, 2015, pp. 4–7).

The present study examines the effect of supplementing a widely used comprehensive whole-child curriculum, the HighScope curriculum, with a skill-based curriculum, *Numbers Plus*, where both were created by the same developer. We first explore perspectives that support comprehensive whole-child instructional approaches, then report on findings from a validation study, which compared children's mathematics outcomes in classrooms that used HighScope to those that supplemented the HighScope curriculum with *Numbers Plus*.

1.1. Comprehensive whole-child preschool instruction

A comprehensive curricular approach that is developmentally appropriate and addresses multiple developmental domains aligns with Head Start's whole-child approach (U.S. Department of Health and Human Services, 2016) and the position held by the National Association of the Education of Young Children (Copple and Bredekamp, 2009). In fact, Component #4 of the Preschool Curriculum Consumer Report (NCQTL, 2015) recommends a comprehensive whole-child approach. Consequently, a large percentage of publicly-funded preschools in the United States use a comprehensive whole-child curriculum, also referred to as a global curriculum. Currently, 73% of Head Start programs, and 41% of state-funded preschools have adopted a global curriculum for their programs, with Teaching Strategies' Creative Curriculum (55% of Head Start; 32% of state pre-k), and the HighScope curriculum (17% of Head Start; 7% of state pre-k) dominating the field (Jenkin & Duncan, 2017).

A theoretical argument for a comprehensive whole-child preschool instruction can be traced back to at least as early as the mid-1800s. The German educator, Friedrich Froebel (1782–1852), along with his contemporaries such as an American educational reformer, Horace Mann (1796–1859), believed that academics is only one factor that explains children's school success. As important are qualities such as moral values and citizenship. Froebel envisioned that the pre-elementary years are essential for developing children's "personality, discipline, and social skills necessary to succeed in school and society" (Jeynes, 2006, p. 1937). Froebel opened his first kindergarten in 1839, which was then introduced to the United States a decade later. Half a century later, a whole-child approach in school continued to be valued.

While the common sense of mankind teaches that the whole nature of the child—body, mind, and heart—should be educated, a common notion has prevailed that the school should educate the mind alone, and, what is worse, the intellect has usually been made so prominent that the feelings and the will have been almost left in abeyance (Petersen, 1893, p. 27).

The argument for a whole-child instructional approach persisted into the 1900s, as evident in articles, such as, *The School's Part in Educating the Whole Child* (N.A., 1933).

In the 1960s, an earlier version of the HighScope curriculum was developed in collaboration with Perry Preschool teachers during the landmark Perry Preschool Project (1962–1967). This set of content and practices that address a full spectrum of children's developmental domains later became known as the Cognitively Oriented Curriculum (Derman-Sparks & Moore, 2016; Weikart, Rogers, Adcock, & McClelland, 1970). Initially influenced by the developmental theory and educational philosophies of Jean Piaget (Piaget & Inhelder, 1969) and John Dewey (1938/1963) Dewey (1938/1963), the HighScope curriculum later took account of the work of Lev Vygotsky (1934/1962) Vygotsky (1934/1962) and expanded its focus on active participatory learning and the process of scaffolding and differentiating children's learning. Throughout the years, HighScope has integrated new research into its curriculum and materials, such as brain research and cognitive, developmental psychology (e.g., Gelman & Brenneman, 2004; National Research Council, 2005; National Research Council, 2009; Shore, 1997; Thompson & Nelson, 2001).

Currently, the majority of the longitudinal evidence of early education is built on comprehensive whole-child approaches, such as the Perry Preschool Study (Heckman & Karakapula, 2019; Schweinhart et al., 2005), Carolina Abecedarian Study (Campbell et al., 2012; Ramey, 2018a, 2018b), and Chicago Longitudinal Study of the Child-Parent Center (Reynolds, Temple, & Ou, 2010; Reynolds et al., 2011). Despite claims that the impact of preschools "fade out," in early elementary school years (e.g., Lipsey, Farran, & Hofer, 2016; Puma et al., 2012), others have explained that non-cognitive skills, such as ability to regulate emotions and behaviors and having a positive outlook in life (e.g., Heckman & Karakapula, 2019), may have persisted and helped participants experience long-term benefits.

The predominance of the whole-child approach in the publicly-funded preschool landscape also stems from Edward Zigler and his colleagues' notion of "social competence" as the desired outcome for Head Start children (Raver & Zigler, 1997; Zigler & Trickett, 1978). When assessing program outcomes, Zigler and his colleagues argued that children's social competence needs to be measured, that is, "the interrelatedness of cognitive and intellectual development, physical and mental health, nutritional needs, and other factors that enable a child to function optimally" (Love, Chazan-Cohen, & Raikes, 2007, p. 85). Recently, Halle and Darling-Churchill (2016) defined social competence as one of four subdomains of social and emotional development, which includes skills such as "interacting positively with peers and adults through cooperation, listening, taking turns, and initiating and maintaining conversations; engaging in social problem solving; understanding the rights of others; treating others equitably; distinguishing between incidental and intentional actions; and balancing one's own needs with the needs of others (p. 9)." Accumulating evidence, especially with young children, supports the importance of a whole-child approach that focuses on social competence. That is, social-emotional and cognitive development are interrelated (e.g., Mackintosh and McCoy, 2019; Raver, 2002), and both are enhanced through warm and competent relationships (Shonkoff and Phillips, 2000). In addition, while early mathematics skills have been found to be the best predictor of later school success (Duncan et al., 2007), early social competence predicts early math skills (Mackintosh and McCoy, 2019).

In recent years, content-specific curricula or targeted curricula, which focus on specific content areas or skills, such as literacy or mathematics, have shown evidence of enhancing children's learning (e.g., Jenkin & Duncan, 2017; Weiland, McCormick, Mattera, Maier, & Morris, 2018). Jenkin and Duncan (2017), for example, go as far to argue that targeted curricula are superior to comprehensive whole-child curricula or global curricula, and that global curricula do not improve children's outcomes any more than "an assortment

of activities that early childhood education centers develop on their own” (p. 40).

The criticism of global curricula comes at a time when academic demands on kindergarteners have increased drastically compared to two decades ago. Bassok, Latham, and Rorem (2016) compared kindergarten teachers' survey responses from 1998 and 2010 and found that kindergarten teachers in 2010 spent more time teaching academic skills like literacy and mathematics than those in 1998, and less time on arts, science, music and movement. Focus on child outcomes becomes pertinent as we ask the question, *how* can we best prepare children for the demands of kindergarten? If early childhood education is to continue to support a whole-child approach, *how* can we accomplish such instruction more effectively given different curricular evidence and options?

Similar questions were raised by the Advisory Committee on Head Start Research and Evaluation (2012) in their final report. While noting that targeted curricula may be more effective than widely used global curricula in preparing children for school, the committee stressed the importance of educating the whole child. An obvious shortcoming of targeted curricula is that they target only one domain, such as literacy, mathematics, or social skills, while the whole-child global curriculum addresses all these domains. The committee, therefore, underscored existing evidence for two options, that is, to target other content areas either by a combination of targeted curricula (e.g., Boston Public Schools Universal Prekindergarten which uses *Building Blocks* for mathematics and *Opening the World of Learning* (OWL) for literacy; Weiland & Yoshikawa, 2013) or by embedding a targeted curriculum in a comprehensive whole-child curriculum (e.g., the Head Start Research-based, Developmentally Informed or REDI Intervention; Bierman et al., 2008). The current study tests the latter approach of supplementing the comprehensive HighScope curriculum, with a targeted HighScope's *Numbers Plus Preschool Mathematics Curriculum* (or *Numbers Plus*; Epstein, 2007).

1.2. Supplementing global curricula with targeted curricula

The Head Start REDI study is one prominent source of evidence for the effectiveness of supplementing comprehensive whole-child curricula with targeted curricula (Bierman et al., 2008). The whole-child curricula used by classrooms in the Head Start REDI study were Creative Curriculum (55%) and HighScope (45%; Domitrovich et al., 2009). Embedded within those curricula were the Promoting Alternative Thinking Strategies or PATH, which are enhanced research-supported practices in language/emergent literacy and social-emotional development (Domitrovich, Cortes, & Greenberg, 2007).

The Head Start REDI was conducted as a part of the Interagency School Readiness Consortium (ISRC; Griffin, 2010) established by several federal agencies in 2003. The goal of ISRC was to build stronger evidence of comprehensive and integrated preschool curriculum and to “move away from piecemeal interventions that focused narrowly on specific skill domains” (Bierman et al., 2014). Forty-four Head Start classrooms from three counties in Pennsylvania were first stratified by factors, such as county location, length of program, student demographics, and center size. The classrooms were then randomly assigned to those receiving and implementing the language/literacy and social-emotional enhancements (treatment) or business-as-usual (control) groups. This method ensured that treatment and control groups were more comparable. Treatment teachers were trained to implement the enhancements. Classrooms from the same center were always assigned to the same condition to avoid treatment contamination. The study found that treatment classrooms were rated as having higher quality adult-child interactions (Domitrovich et al., 2009) and treatment children performed better in vocabulary, emergent literacy, emo-

tional understanding, social problem solving, social behavior, and learning engagement (Bierman et al., 2008). A recent longitudinal follow-up of the Head Start REDI found that children who were in the Head Start REDI classrooms continued to exhibit better social-emotional skill in second grade (Bierman et al., 2008).

While the researchers reported that enhancements were intentionally aligned with the global curricular approaches, qualitative follow-up conducted a year after the Head Start REDI efficacy trial revealed multiple barriers to REDI implementation. These included (Sanford DeRousie & Bierman, 2012, p. 60):

- REDI conflicts with other program requirements (70%)
- Not enough time to do REDI (50%)
- REDI not developmentally appropriate (50%)
- Limited or no support for using REDI (40%)
- REDI conflicts with teachers' philosophy and style (40%)

One teacher is quoted as saying (Sanford DeRousie & Bierman, 2012, p. 60):

It's just hard. We're trying to do everything, we're trying to do High/Scope, we're trying to do themes, and we're trying to do REDI and it's, you can't do everything.

The findings suggest that the blending of PATH into whole-child/parent curricula may not have been as seamless as intended by the Head Start REDI researchers.

1.3. Preschool mathematics curriculum

At least three preschool mathematics curricula have contributed to the evidence of supplementing global curricula with targeted curricula—*Big Math for Little Kids*, *Pre-K Mathematics*, *Building Blocks for Math*. Early mathematics is especially important as early math competence predicts later mastery of complex mathematical skills (Duncan et al., 2007). In addition, students who are successful in math are not only more likely to graduate from high school and complete college, but also to land higher skilled jobs with higher earnings (Reyna, Nelson, Han, & Dieckmann, 2009; Ritchie & Bates, 2013). In 2002 (updated in 2010), the National Association for the Education of Young Children (NAEYC) and the National Council of Teachers of Mathematics (NCTM) jointly outlined the social-economic and racial/ethnic disparity in mathematical achievements among American students which begin in early childhood; low-income children consistently fall behind middle- and upper-income children in developing sufficient early mathematical foundations in preschools and at home (Bowman, Donovan, & Burns, 2001; Denton & West, 2002; Natriello, McDill, & Pallas, 1990; Starkey, Klein, & Wakely, 2004).

Despite the importance of a strong early foundation in mathematics for later school and life success, mathematical learning is rarely emphasized in preschool classrooms (Gerde, Pierce, Lee, & Van Egeren, 2018; National Research Council, 2009; Parks, Wakabayashi, & Hardin, 2018). Many preschool teachers express anxiety with mathematics; that is, they have negative experiences with mathematics growing up, have limited content knowledge in math, and are ill-equipped and uncomfortable introducing and engaging children in mathematical thinking (Clements & Sarama, 2007; Copley, 2004; Gerde et al., 2018; Ginsburg et al., 2006; Lee & Ginsburg, 2007). The joint NAEYC/NCTM statement of 2002/2010, thus, called to action the urgent need for enhanced teacher training and increased quality and quantity of early mathematics instructions in preschool classrooms (NAEYC/NCTM, 2010).

Big Math for Little Kids. The efficacy study of *Big Math for Little Kids* (BMLK) examined whether the targeted early mathematics curriculum is effective “above and beyond the mathematics instruction that typically occurs in preschool and kindergarten”

(Presser, Clements, Ginsburg, & Ertle, 2015, p. 408). Sixteen publicly subsidized child care centers were randomly assigned to treatment (enhancements with BMLK) or control (business-as-usual). Given that the majority of the study classrooms used the Creative Curriculum, the researchers compared selected features of BMLK with the Creative Curriculum. They concluded that while the Creative Curriculum's play-based instructional philosophy, in which teachers are observers, is "promising," there is a need to provide teachers with more explicit strategies (learning goals, planning charts, take-home activities, and assessment) to effectively support children's mathematical learning.

The study found that children who were in the BMLK treatment classrooms scored significantly better on mathematical assessments than those in the control classrooms. In addition, higher percentages of children in the BMLK treatment classrooms were able to accurately identify shapes and explain their responses (Presser et al., 2015). The effect size (Cohen's $d = 0.32$) was moderate, and significant differences in mathematics outcomes were not evident until the second year of the intervention.

Pre-K Mathematics. *Pre-K Mathematics* (Klein, Starkey, Clements, Sarama, & Iyer, 2008; Starkey et al., 2004) was developed to strengthen preschool and home mathematics environments for children. The curriculum consists of both planned activities and pre-selected materials for small-group activities to be used in preschool classrooms, and take-home activities and materials aligned to the classroom activities. The curriculum was tested using a randomized control trial in 40 Head Start and state pre-kindergarten classrooms, which used a variety of "general-purpose" curricula, including Creative Curriculum and HighScope. The classrooms were randomly assigned to treatment (Pre-K Mathematics) or control (business-as-usual). Block randomization was conducted to ensure comparability across a variety of factors between treatment and control groups, and classrooms in the same site were assigned to the same condition to avoid contamination of intervention. Children in treatment classrooms demonstrated significantly higher mathematical knowledge than control children at post-test. The effect size was moderate, based on What Works Clearinghouse calculation (Cohen's $d = 0.58$; Klein et al., 2008).

Building Blocks for Math. *Building Blocks for Math* consists of developmentally sequenced activities in a variety of intervention formats, including small- and whole-group activities and software. It uses the learning trajectories framework; that is, "to attain a certain mathematical competence in a given topic (goal), children learn each successive level of thinking (the developmental progression), aided by tasks (instructional activities) designed to build the mental actions-on-object that enable thinking at each higher level" (Clements & Sarama, 2011, p. 969). The efficacy study of *Building Blocks* (Clements & Sarama, 2007) was conducted at two sites. One site used the Creative Curriculum, hence *Building Blocks* plus Creative Curriculum was tested against business-as-usual Creative Curriculum classrooms. Treatment children demonstrated greater gains in mathematical concepts and skills. The effect size across this and an earlier impact study was large at 1.07 (U.S. Department of Education, 2007).

More recently, the *Building Blocks* curriculum has been scaled up using a TRIAD (Technology-enhanced, Research-based, Instruction, Assessment, and professional Development) framework (Clements & Sarama, 2011; Clements, Sarama, Spitler, Lange, & Wolfe, 2011). An evaluation of the scale-up was conducted with forty-two schools serving low-resource communities. The schools were randomly assigned to treatment (TRIAD) with and without follow-through into Kindergarten/first grade and control (business-as-usual). Due to a stronger focus on mathematics education in the districts, control classrooms implemented a variety of early mathematics curriculum with some incorporating an earlier version of the *Building Blocks*. Despite the higher mathematics pedagogical content

knowledge among control teachers, researchers found significant improvements in the acquisition of mathematical concepts and skills amongst children in treatment classrooms over control classrooms, with a large effect size ($g = 0.72$; Clements et al., 2011). The treatment children with follow-through instructions continued to perform significantly higher in Kindergarten (Sarama, Clements, Wolfe, & Spitler, 2012) and first grade (Clements, Sarama, Wolfe, & Spitler, 2013), in comparison to both treatment children without follow-through and control children. The *Building Blocks* has also been tested in combination with other curricula and found to be effective in improving children's mathematical learning (e.g., Clements, Sarama, & Germeroth, 2016; Weiland & Yoshikawa, 2013).

1.4. Early mathematics in HighScope

Although the comprehensive HighScope curriculum includes mathematics content, the joint call-to-action by NAEYC/NCTM (2002/2010) NAEYC/NCTM (2002/2010) motivated the development of HighScope's skill-based, content-focused *Numbers Plus*. In HighScope classrooms, teachers observe and reflect on children's development in natural settings and plan activities based on children's emerging skills and interests. Each planned activity is tied to one or more of 58 Key Developmental Indicators (KDIs), which serve as HighScope's learning standards. KDIs "are the building blocks of thinking and reasoning at each stage of a child's development" (Epstein, 2014, p. 10) and guide comprehensive development of the whole child. In this way, the teachers participate as partners in children's learning.

Currently, nine of the 58 Key Developmental Indicators (KDIs) refer to mathematical teaching in HighScope classrooms. These KDIs guide teachers' planning of their daily mathematical instruction. As well, HighScope teachers monitor children's development using the COR Advantage (HighScope, 2014; Wakabayashi, Claxton, & Smith, 2017), a teacher-scored observational child assessment system aligned to the KDIs. The mathematics KDIs are (Epstein, 2014, pp.12):

- (1) Number words and symbols: Children recognize and use number words and symbols (KDI #31).
- (2) Counting: Children count things (KDI #32).
- (3) Part-whole relationships: Children combine and separate quantities of objects (KDI #33).
- (4) Shapes: Children identify, name, and describe shapes (KDI #34).
- (5) Spatial awareness: Children recognize spatial relationships among people and objects (KDI #35).
- (6) Measuring: Children measure to describe, compare, and order things (KDI #36).
- (7) Unit: Children understand and use the concept of unit (KDI #37).
- (8) Patterns: Children identify, describe, copy, complete, and create patterns (KDI #38).
- (9) Data analysis: Children use information about quantity to draw conclusions, make decisions, and solve problems (KDI #39).

Each KDI is accompanied by a scaffolding chart which includes (1) a description of the KDI (e.g., for KDI #31, "Children recognize and name numerals in their environment. They understand that cardinal numbers [e.g., one, two, three] refer to quantity and that ordinal numbers [e.g., first, second, last] refer to the order of things. They write numerals."), (2) what children may say or do at different developmental levels (earlier, middle, later), (3) ideas for what adults can do to support children at their developmental levels, and (4) ideas for what adults can do to gently extend the children's knowledge, skills, and understanding. HighScope's COR Advantage (HighScope, 2014), an authentic child observational assessment

Table 1Alignment between NCTM curricular focal points, *Numbers Plus*, and HighScope Key Developmental Indicators (KDIs).

NCTM curricular focal points	<i>Numbers Plus</i>	HighScope KDIs
Numbers and Operations	Numbers and Operations	KDI #31 Number words and symbols KDI #32 Counting KDI #33 Part-whole relationships
Geometry and Spatial Reasoning	Geometry	KDI #34 Shapes KDI #35 Spatial awareness
Measurement	Measurement	KDI #36 Measuring KDI #37 Unit
<i>Areas related to focal points</i>		
Algebra	Algebra	KDI #38 Patterns
Data analysis	Data analysis	KDI #39 Data analysis

conducted by teachers, is aligned to the KDIs. For example, to score level 3 (out of 7) in the category of Number and Counting, “child consistently counts (with one-to-one correspondence) up to 10 objects.” To obtain a score of 5 (out of 7), “Child counts (with one-to-one correspondence) more than 10 objects and says the last number counted tells how many.”

During the 20-day HighScope Preschool Curriculum Course, 1-day is dedicated to training teachers on these nine mathematics KDI. All HighScope teachers are expected to cover these mathematics KDIs during adult-initiated small- or large-group times, as well as throughout the day, and monitor children’s progress in mathematics learning using the COR Advantage. The HighScope curriculum encourages teachers to provide ample opportunities for children’s active participatory learning in mathematics. However, it does not prescribe frequencies, sequence or detailed steps for activities that teachers can implement in their classrooms to target those KDIs. This is because HighScope’s premise is that teachers are trained professionals who intentionally plan their lessons around each child or each classroom’s strengths and needs.

Not all teachers, however, have skills and knowledge to design and deliver effective mathematical learning opportunities for children in their classrooms. In fact, results from recent studies indicate that preschool teachers often have limited content knowledge, training and guidance in engaging children in mathematical thinking (Gerde et al., 2018; Ginsburg et al., 2006; Lee & Ginsburg, 2007; Parks et al., 2018; National Research Council, 2009). *Numbers Plus* was therefore developed to provide preschool teachers with ideas to encourage more frequent and intentional mathematical learning in classrooms and expand children’s mathematics KDIs during adult-initiated group times. Additional training/coaching to implement *Numbers Plus* helps to deepen teachers’ pedagogical content knowledge and expand their repertoire for supporting children’s mathematical development. This article reports on the findings of the validation study, which compared children’s mathematics outcomes in classrooms that used the HighScope curriculum to those who supplemented the HighScope curriculum with *Numbers Plus*.

2. Research questions

Our questions are: (1) Will teachers and coaches implement the *Numbers Plus* curriculum with fidelity? (2) Does supplementing a global HighScope curriculum, with a targeted early mathematical curriculum, *Numbers Plus*, enhance children’s mathematical learning?

We hypothesize that teachers and coaches are able to implement the *Numbers Plus* curriculum with fidelity, and children in classrooms that participate in the HighScope plus *Numbers Plus* curriculum result in higher math outcomes than those in HighScope only classrooms. Our study is the first effort to examine the effect of supplementing a widely used comprehensive whole-child global curriculum with a skill-based targeted mathematics curriculum, where both were created by the same developer using the same

instructional philosophy and approach. The present study hopes to demonstrate a promising and sustainable approach to improve the effectiveness of a global curriculum through content-focused enhancement efforts.

2.1. *Numbers Plus*

Numbers Plus and the HighScope mathematics KDIs align with the standards and focal points of the National Council of Teachers of Mathematics (NCTM; Fuson, Clements, & Beckmann, 2010; see Table 1), and the teaching-learning path identified by the National Research Council’s Committee on Early Mathematics Education (NRC, 2009). While the intent of the HighScope KDIs is to help teachers plan and monitor math content introduced throughout the day, the purpose of *Numbers Plus* is to further guide teachers to conduct focused mathematical lessons during teacher-directed group times.

The *Numbers Plus* kit contains a set of 120 small- and large-group activity cards (104 small-group, 14 large-group, and 2 either small- or large-group) divided into five key content areas: number sense and operations (36 activities), geometry (27 activities), measurement (25 activities), algebra (16 activities), and data analysis (16 activities). The *Numbers Plus* activities were thus designed to focus on the first three curriculum focal points—number sense and operations, geometry, and measurement—with the largest emphasis on number sense and operations (Fuson et al., 2010; NRC, 2009). While not a part of the curricular focal points, algebra and data analysis were included in *Numbers Plus* as related connections (Neill and Gainsley, 2014). This arrangement reflects the NCTM recommendation that “work on patterns and data can be woven into work on the three focal points, but not with the same time investment” (Fuson et al., 2010, p. 1).

Teachers can choose the activity they want to implement based on children’s interests and needs, thus preserving HighScope’s premise that teachers are trained professionals. Each card has suggestions for how to introduce the activity (beginning), engage children (middle), and transition out of the activity (end). It also has a built-in scaffolding chart with suggested adult scaffolding strategies, similar to the KDI scaffolding chart, so children of all developmental and ability levels can participate successfully and have a valuable learning experience. In *Numbers Plus*, therefore, children’s mathematical learning is sequenced within, rather than across, activities (Neill and Gainsley, 2014; see Appendix A for a sample activity card). The curriculum kit also includes a Teacher’s Manual, and English and Spanish versions of a parent handbook to share information about the importance of early mathematics, and strategies for promoting mathematics at home.

The *Numbers Plus* training consists of seven modules. Module one covers Numbers and Operations (KDI #s 31–33) and Geometry (KDI #s 34–35); module two covers Measurement (KDI #s 36–37); module three covers the content areas of Algebra and Data Analysis (KDI #s 38–39); and module four covers strategies to help

teachers use existing classroom materials and resources to integrate mathematics learning throughout the day. For the current validation study, the training was conducted as a combination of face-to-face (6 days) and online training segments, accompanied by 20 follow-up coaching contacts.

Given that *Numbers Plus* was developed by HighScope, and training and coaching are offered by HighScope trainers, we expected continuity between the parent curriculum (i.e., HighScope) and *Numbers Plus*. For example, the HighScope curriculum requires at least one small-group time during the day, into which *Numbers Plus* activities can be easily incorporated. The information on the activity cards (i.e., scaffolding charts) is presented in a format familiar to HighScope teachers. In addition, HighScope teachers use KDIs for planning, and the activity cards indicate how each math activity relates to the KDIs. In other words, the targeted curriculum, *Numbers Plus*, was intentionally designed to align with the HighScope curriculum to seamlessly enhance its use in mathematics instruction.

3. Method

3.1. Study sample

A total of 22 sites, 52 classrooms, and 583 children (323 treatment; 260 control) were initially selected to participate in the study. HighScope early childhood and educational services staff members who are knowledgeable about the amount of HighScope trainings provided, and the levels in which the HighScope curriculum is implemented, were consulted to identify high-quality HighScope programs. All programs that were approached wrote a letter of support for the research grant proposal and agreed to participate once the grant was awarded. The sites were located in three states and data collection was conducted across three years in Michigan (Year 1), Maine (Year 2), and Ohio (Year 3). Attrition was not an issue, following the WWC attrition standards (Institute of Education Sciences, 2017). Comparing the baseline measurement with the post-test measurement, the overall attrition was 10.7 percent–9.9 percent for the treatment group and 11.6 percent for the control group; the differential attrition was 1.7 percent. The overall attrition and the differential attrition are within the ranges that do not represent bias in the estimates (IES, 2017).

Site characteristics. Table 2 shows the site characteristics across Years 1–3. All programs/sites used HighScope as their primary curriculum, and all teachers were expected to use HighScope's key developmental indicators (KDIs) in mathematics and HighScope COR Advantage as tools to observe, describe, and support children's math development. All sites were also staffed with certified HighScope trainers. Michigan sites consisted of mainly tuition-based classrooms located in a suburban context. Maine sites were mainly school-based publicly-funded classrooms in rural locations. Ohio sites were federally-funded Head Start classrooms in an urban city. There was no attrition of sites or classrooms from the study.

Child and family characteristics. Table 3 shows the number of children in treatment or control classrooms, number eligible for the study, number selected, and attrition. Overall, 89 percent of the initially selected sample (90% of treatment and 88% of control) were retained for post-assessment.

Table 4 shows children and parent demographics and background across Years 1–3. Statistically significant differences were found in two demographics in two sites (Michigan and Ohio). In Michigan these significant differences were in ethnicity ($p < 0.001$) and primary language at home ($p < 0.01$) while in Ohio the only significant difference was found in ethnicity ($p < 0.05$).

Additionally, a large effect size was observed in the case of ethnicity, mainly due to empty cells (e.g. in Michigan, the treatment

Table 2
Site characteristics.

Year/State	Setting	Treatment			Control			Total	
		Program Type	#Sites	#Classrooms	#Children Selected	Program Type	#Sites	#Classrooms	#Children Selected
Year 1: Michigan	Suburban	For-profit tuition-based	4	11	111	For-profit tuition-based/1 state-funded PreK classroom	3	7	66
Year 2: Maine	Rural	State-funded PreK/1 Head Start classroom	5	8	95	State-funded PreK/1 Head Start classroom	4	6	81
Year 3: Ohio	Urban	Head Start	3	10	117	Head Start	3	10	113
Total			12	29	323		10	23	260
									583

^a In Year 3, 117 (of 151 children) were randomly selected to be assessed for the treatment group and 113 (of 167 children) for the control group, for a total of 230 children.

Table 3
Child Sample Selection and Attrition.

Year/State	Group	# Classroom	Total # in class	Total # eligible	% Eligible	# Initially selected	# Pre-tested	# Post-tested	% Retained
Year 1: Michigan	Treatment	11	225	136	60%	111	111	109	98%
	Control	7	148	81	55%	66	65	58	89%
	Subtotal	18	373	217	58%	177	176	167	95%
Year 2: Maine	Treatment	8	121	102	84%	95	92	88	96%
	Control	6	99	87	88%	81	81	73	90%
	Subtotal	14	220	189	86%	176	173	161	93%
Year 3: Ohio	Treatment	10	172	151	88%	113	109	84	77%
	Control	10	175	167	95%	117	103	89	86%
	Subtotal	20	347	318	92%	230	212	173	82%
Total		52	940	724	77%	583	561	501	89%

Table 4
Children demographic by site and treatment condition.

Children characteristic		Year 1 (Michigan site)			Year 2 (Maine site)			Year 3 (Ohio site)		
		Treatment	Control	Effect Size	Treatment	Control	Effect Size	Treatment	Control	Effect Size
Mean age in months	Mean	49	50.73	−0.31	53.19	53.43	−0.06	52	51.47	0.09
	SD	(5.51)	(5.53)		(4.26)	(3.77)		(5.87)	(6)	
Gender	Female	50	35.56	0.29	55.22	56.67	−0.03	45.45	59.46	−0.28
Ethnicity/Race	Asian	0	20	−0.93	0	1.72	−0.26	0	0	0
	Black	4.21	2.22	0.11	0	0	0	41.54	67.57	−0.53
	Hispanic	0	0	0	0	0	0	1.54	1.35	0.02
	Other	1.05	0	0.21	0	0	0	0	0	0
	White	84.21	64.44	0.46	94.03	94.83	−0.03	47.69	24.32	0.49
	Multiracial	10.53	13.33	−0.09	5.97	3.45	0.12	9.23	6.76	0.09
English as Native language		94.74	77.27	0.53	98.46	94.92	0.21	90.91	97.26	−0.28
Children with disability		11.46	15.56	−0.12	25.37	31.67	−0.14	27.27	24.32	0.07
Children with IEP		4.17	8.89	−0.19	8.96	13.33	−0.14	4.55	5.41	−0.04
Low births		8.6	4.65	0.16	3.08	3.57	−0.03	8.93	11.94	−0.1
Parents who are married		92.63	88.89	0.13	62.69	56.67	0.12	36.92	26.03	0.24
Parent education	No HS diploma	2.11	0	0.29	9.09	6.78	0.09	25.76	21.62	0.1
	HS diploma/some college	6.32	8.89	−0.1	54.55	64.41	−0.2	57.58	63.51	−0.12
	Associate	3.16	0	0.36	9.09	6.78	0.09	6.06	8.11	−0.08
	Bachelor's or above	88.42	91.11	−0.09	27.27	22.03	0.12	10.61	6.76	0.14
Parent employed		85.42	77.78	0.2	74.24	66.67	0.17	55.38	59.46	−0.08
Public assistance	Food Stamps	3.12	6.82	−0.17	37.31	42.37	−0.1	72.31	68.49	0.08
	TANF	1.05	0	0.21	14.06	12.07	0.06	15.87	21.13	−0.14
	Medicaid	4.17	6.67	−0.11	45.45	51.67	−0.12	71.21	76.06	−0.11
	Free lunch	3.16	4.55	−0.07	37.5	47.46	−0.2	70.97	70.42	0.01
Wealth index	Mean	0.73	0.68	0.13	−0.08	−0.19	0.19	−0.58	−0.62	0.1
	SD	(0.38)	(0.41)		(0.6)	(0.58)		(0.49)	(0.4)	

group had no Asian children). Other differences were found in children's native languages. In the case of Michigan, more children in the treatment group had English as the native language (0.53). Nevertheless, most of the effect sizes were small, with no systematic pattern showing imbalances between groups across all three sites.

Eligibility for participation in the project was determined by children's English language comfort level and age. English language comfort became one of the criteria because of our choice to use the Early Math Assessment System (EMAS; Ginsburg, Pappas, Lee, & Chiong, 2010) as one of the preschool math assessments. Given that EMAS was only available in English, we decided not to test children whose most comfortable language was a language other than English. Fewer than five children were screened out by parents during the consent process or by their teachers. An attempt to assess a child who was unable to respond in English was made once in Year 1, but the child was subsequently identified by the teacher as being an English language learner (ELL) and so screened out on that basis.

In addition to the issue of language, EMAS, at the time, was set up so that its 3-year-old, 4-year-old, and 5-year-old assessments were different and were not scored on a continuous scale. To facilitate the process of testing the same children pre- and post-intervention with the same assessment, we decided to use only the 4-year-old assessment for this project. This decision meant that children who

were too young for the assessment were ineligible to participate. The upper-age cutoff was set based on Michigan's kindergarten entry cutoff date, which was December 1 at the time the project was initiated. Thus, participants were selected from those born between 12/2/2006 and 5/31/2008 for Year 1; 12/2/2007 and 5/31/2009 for Year 2; and 12/2/2008 and 5/31/2010 for Year 3.

Teacher characteristics. There was no difference found between treatment and control group teachers' education reported at the beginning of the year across Years 1–3. Seventy-eight percent of the treatment teachers and 81 percent of the control teachers had at least an associate degree in an early childhood-related field. There was also no difference in teachers' experiences. For both treatment and control groups, 68 percent of the teachers in the combined sample reported having six or more years of experience in the early childhood field. In Years 1 and 2, a large majority of the teachers were white (93% in Year 1; 100% in Year 2). In Year 1, 7 percent were black, and 5 percent were also Hispanic/Latino. In Year 3, 55 percent were white and 43 percent were black (2% did not answer). Five percent were Hispanic/Latino. There was no difference in teachers' race or ethnicity across treatment and control groups.

Across the three years, six treatment teachers (three of them lead teachers) and seven control teachers (two of them lead teachers) left in the middle of the program year. In HighScope classrooms,

lead and assistant teachers co-teach so that both are given equal responsibilities and interact similarly with children. Given that both lead and assistant teachers in the treatment group were trained in *Numbers Plus*, at least one trained teacher remained in each participating treatment classroom, and at least one classroom with a trained lead teacher remained per site. Overall, 89 percent of the treatment lead teachers and 91 percent of the control lead teachers were retained in the study. Percent of lead teacher attrition by treatment and control group was not significantly different.

3.2. Procedure

Training. Treatment teachers received the *Numbers Plus* training once we completed pre-intervention data collection. The first two of the seven *Numbers Plus* modules were delivered face-to-face, for a total of six days. Aligned with the recommendations of the NCTM (2006) and NRC (2009), teachers were trained initially on Numbers and Operations (KDI #s 31–33) and Geometry (KDI #s 34–35) during module one, and were encouraged to try out several activities in those content areas prior to returning for the second module. Module two included the content area of Measurement (KDI #s 36–37). The first module took place in late fall and the second module in early winter.

The third and the fourth modules were delivered as online training segments. Module three covered the content areas of Algebra and Data Analysis (KDI #s 38–39). Module four included strategies to help teachers use existing classroom materials and resources to integrate mathematics learning throughout the day. Teachers completed module three within four weeks in late winter and module four within three weeks in early spring. Control teachers were offered the same *Numbers Plus* training following post-data collection.

In Year 1 (Michigan), two HighScope Early Childhood Specialists served as trainers. These Early Childhood Specialists were highly knowledgeable and experienced in the HighScope curriculum, and were involved in the development and initial testing of the *Numbers Plus* curriculum. In Years 2 (Maine) and 3 (Ohio), local site supervisors served as trainers. The trainers attended four days of face-to-face *Numbers Plus* Training of Trainers at the HighScope Educational Research Foundation in Ypsilanti, Michigan over the summer prior to implementation. The trainings were facilitated by HighScope Early Childhood Specialists who were trainers/coaches in Year 1. All trainers were HighScope certified trainers, and had at least a bachelor's degree in early childhood education.

Coaching. *Numbers Plus* trainers also served as coaches. In addition to the four days of *Numbers Plus* Training of Trainers, the trainers/coaches attended three days of training on coaching methods facilitated by HighScope Early Childhood Specialists who were trainers/coaches in Year 1. The coaching training took place adjacent to Training of Trainers. *Numbers Plus* coaching strategies were adapted from Hegland and Oesterreich (2011). Each treatment classroom was assigned to a coach who kept detailed records of the contacts they had with each teacher. The coaches observed and assessed teachers' implementation of a *Numbers Plus* activity; discussed with teachers the results of the assessment, teacher strengths, and opportunities for growth; and developed a coaching plan. This coaching plan included a variety of coaching activities including (but not limited to):

- The coach videotaped the teacher and the teacher self-reflected
- The coach videotaped the teacher; the coach and teacher then viewed the video together
- The teacher viewed another teacher's video and reflected on that teacher's strategies
- The coach and teacher watched a training video together

- The coach and teacher viewed another teacher, discussing relevant strategies used
- The coach assigned the teacher to do further reading and reflect on it
- The coach and teacher highlighted strategies to use on a *Numbers Plus* activity card
- The coach and teacher made visual cues based on a *Numbers Plus* activity card
- The coach modeled a *Numbers Plus* activity for the teacher; the two met and discussed the activity afterward
- The coach provided over-the-shoulder coaching to the teacher as he or she conducted a *Numbers Plus* activity
- The coach asked the teacher to do a *Numbers Plus* activity and focus on using a specific scaffolding strategy
- The coach modeled math scaffolding strategies during other parts of the classroom day

After each coaching session, the teacher and the coach discussed their goals and how to attain them. The coach, then, completed a coaching log, which captured his or her thoughts about the teacher's understanding and motivation, the coaching strategy used, what went well, what was challenging, observations, and ideas for the next coaching visit. When a teacher made progress toward his or her goals, the coach reassessed or developed another coaching plan with the teacher. Coaches planned to visit each classroom once every two weeks and initiate up to 20 contacts per classroom, based on the developers' recommendations. There were two coaches in Michigan (Year 1) and Ohio (Year 2) and one coach in Year 3 (Maine). In Year 1, one coach fell gravely ill mid-year and was replaced by a HighScope-contracted trainer.

Data collection. We collected two types of data. The first set of data was collected to evaluate the implementation of *Numbers Plus*, using three measurement tools. We used the Program Quality Assessment (Preschool PQA; HighScope, 2003a) to assess baseline equivalence in the quality of HighScope implementation between treatment and control classrooms. The Preschool PQA (Form A—classroom quality) was completed by trained and reliable PQA assessors at the beginning of each study year.

To assess whether the developers' recommended dosage of *Numbers Plus* was established during the development phase (twice a week) and the number of coaching contacts (20 per classroom) were feasible for generating impact, we used activity logs and coaches' logs. Activity logs were collected from teachers throughout the study year, and coaches' logs were collected from coaches after each coaching contact.

The second set of data was collected to evaluate the effectiveness of implementing *Numbers Plus* in moderate- to high-quality HighScope classrooms. To do so, we collected a variety of data for use in the final multilevel analysis examining the impact of the intervention on children's mathematics outcomes. To examine treatment and control group differences in child-level outcomes at the end of their preschool year, we assessed children's mathematical skills and knowledge at the beginning and end of the program year using two early mathematics assessments widely used in early childhood research. The Woodcock-Johnson III (WJ-III; Woodcock, McGrew, & Mather, 2001) subtest 10—Applied Problems was administered by trained data collectors as the first test for all children. The Early Mathematics Assessment System (EMAS; Ginsburg et al., 2010) was administered at a second sitting. Child outcomes measures were limited to mathematics given the goal of the study, which was to examine the impact of *Numbers Plus* on children's mathematical learning.

PQA and child outcomes measures were collected by data collectors with a bachelor's degree or higher, with experience teaching and/or assessing young children. Data collectors were interviewed over the phone. Those most qualified were then invited to attend

a paid training. To be formally hired, data collectors were required to demonstrate reliability and conduct a practice assessment.

3.3. Measurement

Preschool Program Quality Assessment Form A (Preschool PQA; HighScope, 2003a). The Preschool PQA is a classroom observation tool. Assessors use observational notes and staff interviews to score items on a 5-point scale across four domains: learning environment, daily routine, adult-child interaction, and curriculum planning and child assessment. For each item, there are descriptors that guide assessors' scoring decisions. While the Preschool PQA is designed to be used in any classroom, regardless of curriculum, and used throughout the state of Michigan, it is also an instrument to assess the implementation of HighScope-recommended best practices for HighScope certification. The purpose of using the Preschool PQA for this study, therefore, was to ensure that all classrooms were implementing HighScope at a similar level of fidelity and quality. The instrument's psychometrics report indicates that the Preschool PQA generates reliable and valid scores (HighScope, 2003b). Its interrater reliability ranged from .57 to .75, and Cronbach's alpha averaged .93. The Preschool PQA scores correlated positively with the ECERS (Early Childhood Environment Rating Scale), a similar quality assessment, at .86. Confirmatory Factor Analysis of the four-domain structure was conducted, with factor loading ranging from .43 to .82, with the majority loading at .6 or higher. Preliminary results show moderate but significant correlations with teacher-assessed child outcomes.

Activity logs. Activity logs were collected from treatment group teachers to monitor treatment classrooms' implementation of *Numbers Plus*. Teachers were asked to fill out an activity log, developed for this study, every time they conducted a *Numbers Plus* small-group activity. The completed activity form was used to count the frequency of implementation per classroom as well as the frequency of *Numbers Plus* activities and mathematical topics chosen. The coaches encouraged treatment teachers to conduct at least two *Numbers Plus* small-group times per week, based on the developers' recommendations for what was feasible to implement.

Coaches' logs. The coaches completed a coaching log, after each coaching contact (up to 20 per classroom), which captured his or her thoughts about the teacher's understanding and motivation; the coaching strategy used; what went well; what was challenging; observations; and ideas for the next coaching visit. The completed coaching log was used to count the frequency of coaching contact per classroom to monitor dosage.

Woodcock-Johnson III Tests of Achievement (WJ-III ACH) Applied Problems (subtest 10). WJ-III Applied Problems was used to assess children's ability to analyze and solve practical mathematics problems. For example, children were shown pictures and asked to count the number of items in the picture, identify which picture has the number of items specified, and respond to how many are left if a certain number of items are removed. The average assessment time was 5 min. The instrument has an overall evidence of internal consistency ($>.80$ for all scales), and high median test reliability statistics (Median $r_{11} = .93$, Median SS = 4.08 for Applied Problems). The psychometric report for the mathematics composite of WJ-III ACH (McGrew & Woodcock, 2001) also indicate strong concurrent validity with existing measures such as the Wechsler Individual Achievement Test (WIAT; .56–.70) (Wechsler, 1992) and the Kaufman Test of Educational Achievement (KTEA; .41–.66) (Kaufman & Kaufman, 1985).

Early Mathematics Assessment System (EMAS). Second, EMAS was used to assess children's mathematical knowledge and skills in Numbers/Operations and Shape/Space/Patterns. EMAS was chosen because of its ability to capture the broad range of topics/concepts that children informally acquire in early years. The content tested

also aligned with eight of the nine HighScope KDIs (all except #39 Data Analysis). The context of the assessment is designed around a bear's birthday party and the data collector asks questions that are within the context of playing games at that party. Examples of questions in the Numbers/Operations section of the EMAS include: (a) asking the child to count to a really big number because that is how many presents the bear will get that day, (b) showing the child two numbers and saying the child receives the first number of presents and the bear receives the second number of presents, then asking which number is more, and (c) showing the child some chips, covering them, then adding or subtracting chips while they are covered and asking the child to put the same number of chips under a card. Examples of questions in the Shape/Space/Patterns section of the EMAS include: (a) asking what shape the bear is holding in the picture, (b) asking the child to look at a picture with a pattern on it and then with the picture covered, asking the child to replicate the pattern using the cards given to them, and (c) asking the child to copy how the chips are placed on a board separated into squares. The average assessment time for the Numbers/Operations section of EMAS was 20 min, and the average assessment time for the Shape/Space/Patterns section was 14 min. The EMAS we used was a preliminary version of the Birthday Party Assessment (BP; Ginsburg & Pappas, 2016). Lee (2016) reports on the reliability of the current version as having overall Cronbach's alpha coefficients that range from .70 to .94, and an inter-rater reliability of $>.90$. Concurrent validity with the Young Children's Achievement Test (YCAT; Hresko, Peak, Herron, & Bridges, 2000) was significant, ranging from .37 to .73 for age four.

3.4. Design

The study design is a multi-site cluster randomized trial (MSCRT) consisting of four levels: children, classrooms, sites, and states. Children were nested within classrooms, which were nested within sites, and sites were blocked by states (Spybrook, 2012; Spybrook & Raudenbush, 2009). The states served as natural blocks, that is, data collection at each state was conducted in different study years, and each state consisted of different types of programs (see Table 2). Randomization by site was conducted each year, prior to data collection, once sites were identified. Only sites that implemented the HighScope curriculum with fidelity, that is, ensured that teachers were formally trained in the curriculum and employed certified HighScope trainers, were invited to participate. Randomization was performed at the site level to reduce contamination.

Covariates for impact analysis. The final models included the following covariates: Pretests corresponding to each outcome using the same instruments as the baseline, child age measured in months, and two dummy variables. The first dummy variable was a reference state (Michigan) and the second was the wealth index. Child age in months was used to control for children's developmental differences. The reference state (Michigan) was used to adjust the analysis for differences across sites (Michigan, Maine, and Ohio). The wealth index was derived to adjust for socioeconomic differences across sites.

We created the wealth index using Confirmatory Factor Analysis for categorical data under the assumption that as the level of index increases, more family wealth is reflected in the household. We used income, parental education, employment status, as well as indicators for whether or not the family received public assistance (food stamps, TANF, Medicaid assistance, and free lunch). WLSMV was the estimator used for the factor analysis; fit indices were excellent; Root Mean Square Error of Approximation (RMSEA) was 0.057 with a 90% CI between 0.032 and 0.083; Tucker Lewis Index (TLI) was 0.995; and Comparative Fit Index (CFI) was 0.995. Cut off scores to determine a good fit are suggested as above 0.95

for CFI and TLI and below 0.05 for RMSEA with a 90% CI interval between 0 and 0.1. The Chi-square test of model fit was 33.7 (d.f. = 14, and $p < 0.05$). Loadings for all indicators were above 0.5. Expected income, parental education and parental employment had positive loadings while all the public assistance indicators had negative loadings. The residual variances were less than .4 except for employment status (.78). The wealth index is in standard deviations units, that is, as the index score increases it reflects more wealth.

4. Analysis

4.1. Implementation

Classroom quality. To examine baseline equivalence between treatment and control classrooms, we first compared classroom quality scores for all four areas of the Preschool Program Quality Assessment: learning environment, daily routine, adult-child interaction, and curriculum planning and assessment. Scores were compared using a t -test for independent samples.

Numbers Plus activities. To examine whether teachers implemented *Numbers Plus* at a frequency intended by the developers, we calculated the average number of *Numbers Plus* small-group activities conducted per classroom and per teacher. Teachers were asked to conduct *Numbers Plus* small-group time activities at least twice a week, based on the developers' recommendation.

Coaching contacts. To examine whether coaches implemented their coaching plans as intended by the developers, we summed the number of group meetings and individual contacts with a teacher at the classroom level. We then compared the total number of classroom-level coaching contacts with 20, the developer-recommended frequency of contacts per classroom.

4.2. Impact on child mathematics outcomes

Given that the treatment was randomly assigned to sites while outcomes were measured at the child level, a widely used approach is to use multilevel analyses or hierarchical linear models (HLM) to accommodate the nested structure of the data (levels: children, classroom, sites and states). However, given that there were only three states and very few classrooms per preschool it was not possible to accommodate a 4-level multilevel model; rather 2-level hierarchical linear models with random intercepts were estimated to measure the effects of *Numbers Plus* on children's mathematics outcomes (Gelman & Hill, 2007; Raudenbush & Bryk, 2002; Snijders and Bosker, 2012). All analyses were performed in R using the LME4 package (Bates et al., 2019).

5. Results

5.1. Implementation

Classroom quality. Based on HighScope's Preschool PQA scores at baseline, we concluded that treatment and control classrooms were similar in quality at the outset of the study. No difference was found in the Preschool PQA scores across treatment and control classrooms by year/state, also indicating that classrooms implemented the HighScope curriculum at similar levels of fidelity. This was to be expected, since all the programs had been using the HighScope curriculum for over five years. Michigan sites had the highest average Preschool PQA scores, followed by Ohio sites, then Maine sites (Table 5). An overall score of 3.5 or above is considered moderate to high quality/fidelity.

Numbers Plus activities. The frequency with which teachers implemented *Numbers Plus* small-group activities was consis-

Table 5

Baseline program quality and child outcomes scores.

	Treatment		Control		F-value	t-value
	Mean	SD	Mean	SD		
PQA						
Year 1: Michigan	4.62	0.23	4.47	0.37	1.22	n.s.
Year 2: Maine	3.72	0.41	3.93	0.41	0.89	n.s.
Year 3: Ohio	4.07	0.38	4.04	0.41	0.02	n.s.
Pretests						
<i>WJ-III Applied Problems</i>						
Year 1: Michigan	413.42	16.33	410.86	19.47	−0.85	n.s.
Year 2: Maine	405.68	19.02	405.99	18.38	0.10	n.s.
Year 3: Ohio	384.39	23.75	378.41	22.04	−1.72	n.s.
<i>EMAS</i>						
Year 1: Michigan	23.52778	6.955746	23.48	7.52	−0.04	n.s.
Year 2: Maine	21.75904	6.750954	23.41	6.64	1.54	n.s.
Year 3: Ohio	15.09756	6.19759	15.75	6.33	0.67	n.s.

tent to moderately consistent with developers' intentions, across years and sites. In Year 1, the average number of *Numbers Plus* small-group activities conducted per classroom between November/December of 2011 (post-training period) and April/May 2012 (post-assessment period) was 84.4, with some classrooms conducting as few as 35 and others as many as 176 activities. The wide range in this number can be attributed to the difference in the number of participating teachers per classroom. While all classrooms began the project with 2–3 teachers per classroom, with teacher attrition and maternity leave, three classrooms ended up with fewer trained teachers implementing *Numbers Plus*. On average, each teacher conducted 42 small-group activities, amounting to about 7 *Numbers Plus* small-group activities per month. This number is consistent with the developers' intended frequency goal of 8 *Numbers Plus* small-group activities per month.

In Year 2, the average number of *Numbers Plus* small-group activities conducted per classroom between November/December of 2012 (post-training period) and April/May 2013 (post-assessment period) was 87.4, ranging from 72 to 108. On average, each teacher conducted an average of 44 small-group activities (2 more than Year 1), thus a little more than 7 *Numbers Plus* small-group activities per month.

In Year 3, the last year of implementation, the average number of *Numbers Plus* small-group activities conducted per classroom between November/December of 2013 (post-training period) and April/May 2014 (post-assessment period) was 79.8, ranging from 62 to 89. On average, each teacher conducted an average of 40 small-group activities, averaging about 6.6 *Numbers Plus* small-group activities per month. This was slightly lower than the intended frequency of 8 *Numbers Plus* small-group activities per month.

Teachers also indicated the math content covered in their activity logs. The two areas that *Numbers Plus* indicates as important, and hence teachers are trained on first, were Numbers and Operations and Geometry. We found that, consistent with this training approach, the most popular content areas delivered by teachers during their small-group activities were Numbers and Operations, followed by Geometry (see Table 6).

Coaches' contacts. On average, in Year 1, classrooms had an average of 17.9 recorded coaching sessions completed ($SD = 7.8$). This number is relatively close to the initial design of 20. One reason the number was lower than planned was because one of our coaches fell seriously ill during the project year and had to be replaced by a substitute coach. As a result, the classrooms of this coach did not receive coaching for an entire month while we searched for and trained the substitute coach. In Year 2, classrooms had an average of 22.9 recorded coaching sessions completed ($SD = 4.4$), with all classrooms achieving at least 20 coaching con-

Table 6

Small-group time activities reported by treatment teachers.

	Michigan (Year 1)	Maine (Year 2)	Ohio (Year 3)
Mean number of small-group activities conducted per classroom	84.4 (SD = 43.7)	87.4 (SD = 11.9)	79.8 (SD = 8.5)
% Numbers and Operations	38.8 (SD = 9.8)	30.4 (SD = 2.1)	29.4 (SD = 3.9)
% Geometry	18.6 (SD = 6.2)	25.2 (SD = 3.4)	30.2 (SD = 3.3)
% Measurement	16.1 (SD = 3.8)	16.2 (SD = 5.4)	17.8 (SD = 5.7)
% Algebra	14.4 (SD = 3.2)	15.2 (SD = 3.0)	12.7 (SD = 3.4)
% Data Analysis	12.1 (SD = 2.1)	12.9 (SD = 4.2)	9.9 (SD = 3.6)

Table 7

Multilevel model for EMAS.

	Effect size	Std. error	t-Value	
(Intercept)	−0.59	0.31	−1.91	
Treatment	0.14	0.06	2.34	*
EMAS math pretest	0.56	0.04	15.26	***
Children age	0.01	0.01	2.00	*
Poverty index	0.29	0.06	4.65	***
Maine (reference: Michigan)	0.09	0.09	0.97	
Ohio (reference: Michigan)	−0.32	0.11	−2.88	**

Sample: 400 children nested in 22 sites.

*** $p < 0.001$.** $p < 0.01$.* $p < 0.05$.

tacts. In Year 3, classrooms had an average of 25.4 coaching contacts ($SD = 3.5$). Only one classroom did not reach 20 coaching contacts (i.e., 18); but on average, the number exceeded the expected 20.

Overall, the frequency of *Numbers Plus* small-group times, topics covered, and coaching contacts reached the intended dosage set by the developers, indicating feasibility of implementation.

5.2. Impact on child mathematics outcomes

A series of paired t-tests demonstrates that children in both treatment and control classrooms significantly increased their mathematics knowledge and skills during their preschool year. WJ-III scores increased for control children ($t(218) = -12.62, p < .001$) and treatment children ($t(279) = -16.67, p < .001$). Likewise, EMAS scores increased for both control children ($t(207) = -12.68, p < .001$) and treatment children ($t(269) = -19.91, p < .001$).

The multilevel model on the EMAS post-test (displayed in Table 7), shows significant effects of the treatment. After holding all covariates constant in the model, the small statistically significant effect size difference between children exposed to *Numbers Plus* was 0.14 ($p < 0.05$). No significant differences were found for WJ-III Applied Problems, however, with a very small effect size (-0.02).

6. Discussion

The present study provides preliminary evidence for supplementing a global HighScope curriculum with *Numbers Plus*. While all children increased their mathematics knowledge and skills during their preschool year, we found an additional positive effect of *Numbers Plus* on children's math outcomes, as measured using EMAS. However, the effect size was small compared to existing targeted mathematics curricula which have been moderate (*Pre-K Mathematics*) to large (*Building Blocks*).

The small effect size may be explained by the fact that control classrooms were implementing the HighScope curriculum, which already includes a mathematics component, with moderate to high fidelity, as indicated by the Preschool PQA scores (see Table 5). In previous studies which used HighScope as one of the control group curricula (e.g., Head Start REDI), classrooms' degree of fidelity to the curriculum was unclear. Hence it was difficult to decide if the

effect found was above and beyond global curricula implemented with quality.

The effect size may have been small because the intervention was for only one school year. In addition, given that the training took place through early spring, teachers may not have had sufficient time to fully implement *Numbers Plus*. The efficacy study for BMLK (Presser et al., 2015) also reports that the impact on child outcomes was only observed in the second year of the intervention (kindergarten), but not in the first (preschool).

The impact of *Numbers Plus* was found with EMAS but not with WJ-III Applied Problems. The findings may suggest that *Numbers Plus* was more effective in improving children's mathematical skills but not necessarily a more global mathematical problem-solving ability. Alternatively, the WJ-III Applied Problems may not be sensitive enough to assess changes in young children's mathematical problem-solving skill. Klein et al. (2008), for example, note that WJ-III Applied Problems is "inadequate for the range of mathematical knowledge that is developing during early childhood."

There is, of course, a possibility that the small effect size found in this efficacy study, in combination with significant impact for EMAS but not for WJ-III Applied Problems, imply that *Numbers Plus* is not as effective as the other mathematics curricula previously studied. We address implications and future directions in the following sections.

7. Implications

The findings from this study imply that there is promise in *Numbers Plus* as an evidence-based early mathematics supplement to the whole-child comprehensive HighScope curriculum. However, further examination of why the effect size found was small, and areas in need of improvement (e.g., coaching and training, content covered, or implementation) are left for future work. Finding evidence of effectiveness, however, is only the initial step in curriculum evaluation work. Sarana et al. (2012), for example, refer to *persistence* and *sustainability*, where *persistence* is how long the impact of the intervention lasts in individuals and *sustainability* is how long the intervention continues to be implemented with fidelity. The latter issue may, therefore, be worth a brief discussion.

Evidence from the Head Start REDI study suggests that sustaining implementation of a supplementary curriculum could be challenging if teachers perceive it as additional work and misaligned with their main instructional/curricular philosophy (Sanford DeRousie & Bierman, 2012). In the case of the present study, the developers for the targeted curriculum were in the same research center as the developers of the global/parent curriculum. Because the developers had a good command of the parent/global curriculum (HighScope), the supplementary targeted curriculum (*Numbers Plus*) was naturally designed and delivered to fit within the framework of the parent/global curriculum. In addition, the advantage is that the training and any implementation follow-up come from the same place. Consequently, one of the successes that came out of this project was that the *Numbers Plus* curriculum and training are now included in the comprehensive foundational HighScope curriculum training. Together with the online *Numbers Plus* training approximately 300 coaches and 1602 teachers have been

trained since 2011. However, whether *Numbers Plus* is indeed better sustained post-training/coaching is left for future work.

8. Limitations and future directions

There are a few limitations to this study. First, dual-language learners, children with Individualized Education Plans (IEPs), children younger than three years three months at the start of the school year, and children older than the kindergarten cut-off age for the following school year were excluded from the study. Consequently, we cannot generalize results to all preschool children who may benefit from a supplementary mathematics curriculum.

Second, while we included a variety of types of programs to test the implementation of *Numbers Plus* (private tuition-based, publicly-funded/non-Head Start, and Head Start), the program type was confounded with state and study year. Trainers and coaches were different by state/study year. For example, teachers were trained and coached directly by HighScope Early Childhood Specialists in Year 1. Teachers in Years 2 and 3, on the other hand, were trained and coached by HighScope trainers affiliated with each site. In addition, although Year 1 (Michigan) focused on private tuition-based preschools in suburban neighborhoods, there were two classrooms within two sites which served children through a state-funded preschool grant. Consequently, while it is promising that significant math gains were observed each year, we cannot generalize or compare results by types of programs.

Third, *Numbers Plus* effect sizes were small, and we recommend additional follow-up studies. Unlike *Building Blocks* which has large effect sizes, *Numbers Plus* activities are, for the most part, not sequenced. However, some are sequenced. Currently, 19 (16%) of the 120 activity cards suggest that teachers try an earlier activity or introduce children to certain skills or knowledge. While more research is needed to address the impact of the present version of *Numbers Plus*, another route for HighScope may be to test whether adding more guidance to sequence the 120 activity cards in *Numbers Plus* per expert recommendations may lead to improved mathematics learning.

Fourth, we mentioned earlier that one of the successes of this project was that the *Numbers Plus* curriculum and training are now included in the comprehensive foundational HighScope curriculum training. Research is needed to assess whether the ability to incorporate training into the foundational curriculum training indeed improves continuity and curriculum effectiveness.

Fifth, the frequency of *Numbers Plus* small-group activities that the teachers conducted and the number of coaching contacts per classroom were consistent with the intended dosage specified by the developers. While this study shows that the recommended dosages are feasible to implement in diverse classrooms, the current study did not attempt to capture how teachers implemented the recommended *Numbers Plus* instructional practices/approaches. Preliminary results of the fidelity criteria coding, using small-group-activity video recordings, indicate that there is variability in the quality of *Numbers Plus* implementation, an observation also noted by Presser and others (2015) for BMLK.⁶ Follow-up analysis is currently underway to better assess *Numbers Plus* fidelity of instructional practices/approaches. Once we are better able to identify teachers' fidelity in a deeper way, further study is recommended to examine whether increasing such fidelity actually impacts children's mathematics outcomes further.

Lastly, we are in the process of coding and analyzing *Number Plus* small-group-activity video recordings to capture fidelity in a more nuanced way. Such analysis may also help identify effective

math teaching strategies that predict children's higher mathematical growth. The *Numbers Plus* curriculum kit includes ideas for large-group activities and the *Numbers Plus* training emphasizes integrating mathematics throughout the day. Hence implementation during times other than small-group-activity needs to be assessed in future work.

9. Conclusions

Recent research argues that a skill-based, content-specific curriculum may be more effective in raising children's content knowledge and skills in that content area than a whole-child comprehensive curriculum. The findings from the present study contribute to a piece of the larger question of how. That is, is implementation of a combination of evidence-based, skill-based curricula that cover diverse content areas more effective than or as effective as that of a comprehensive whole-child curriculum supplemented by evidence-based, skill-based, content-specific sub-curricula? While our findings are not sufficient to fully address this question, they certainly do suggest that a whole-child, comprehensive curriculum may be strengthened and supplemented to enhance child outcomes. While teachers need to practice an innovation to implement it with fidelity, they are often resistant to additional demands (Sanford DeRousie & Bierman, 2012). When developers of global curricula can readily integrate the content of their new targeted curriculum into the parent curricula and into their menu of professional development, the burden for teachers may be alleviated. An added implication, then, may be that content-focused enhancements may be the future direction for whole-child, comprehensive curriculum developers, like HighScope, to strengthen curricular effectiveness and promote sustained implementation. Partnering with developers of existing evidence-based skill-based curricula for a more coordinated early childhood curricular system may also be a logical option. An alternative, of course, is to make improvements directly to whole-child curriculum development.

On a larger note, the socio-historical context in which children develop is not static, but rather changes due to advances in technology, focus on academics, and attitudes toward diversity and inclusivity. Research may have solidified the evidence for benefits of early childhood education, but curricula or our understanding of effective curricula need continuous scrutiny to account for the slowly changing world in which young children thrive. Even as evidence compounds for targeted curricula, the early childhood field still values and caters to developing the whole child. To ensure that all children are equipped with the social competence needed for success in school and in life, integrating a targeted content-specific curriculum into a parent comprehensive global curriculum may be one promising way forward for enhancing the current curriculum field.

Authors contributions

Tomoko Wakabayashi: Methodology, Writing-Original Draft, Investigation, Supervision, Project administration

Fernando Andrade-Adaniya: Methodology, Formal Analysis, Writing-Review and Editing, Validation

Lawrence J. Schweinhart: Conceptualization, Writing-Review and Editing, Supervision, Funding Acquisition

Zongping Xiang: Conceptualization, Methodology, Data Curation, Funding Acquisition

Beth A. Marshall: Conceptualization, Resources, Writing-Review and Editing, Supervision, Project administration, Funding Acquisition

⁶ Notes and analysis submitted by Bailey, D.E., 2015.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at <https://doi.org/10.1016/j.ecresq.2020.04.002>.

References

- Advisory Committee on Head Start Research and Evaluation. (2012). *Final report*. Washington, DC: U.S. Department of Health and Human Services. https://www.acf.hhs.gov/sites/default/files/opre/eval_final.pdf
- Bassok, D., Latham, S., & Rorem, A. (2016). Is kindergarten the new first grade? *AERA Open*, 2, 1–31. <http://dx.doi.org/10.1177/2332858415616358>
- Bates, D., Mächler, M., Bolker, B., Walker, S., Christensen, R. H., Singmann, H., et al. (2019). *Linear mixed-effects models using 'Eigen' and S4*. R package lme4 v1.1–20. Repository CRAN. <https://cran.r-project.org/web/packages/lme4/lme4.pdf>
- Bierman, K. L., Domitrovich, C. E., Nix, R. L., Gest, S. D., Welsh, J. A., Greenberg, M. T., et al. (2008). Promoting academic and social-emotional school readiness: The Head Start REDI program. *Child Development*, 79, 1802–1817.
- Bierman, K. L., Nix, R. L., Heinrichs, B. S., Domitrovich, C. E., Gest, S. D., Welsh, J. A., et al. (2014). Effects of head start REDI on children's outcomes 1 year later in different kindergarten contexts. *Child Development*, 85, 140–159.
- Bowman, B. T., Donovan, M. S., & Burns, M. S. (Eds.). (2001). *Eager to learn: Educating our preschoolers*. Washington, DC: National Academy Press.
- Campbell, F. A., & Ramsey, C. T. (1994). Effects of early intervention on intellectual and academic achievement: A follow-up study of children from low-income families. *Child Development*, 65, 684–698.
- Campbell, F. A., Pungello, E. P., Burchinal, M., Kainz, K., Pan, Y., Wasik, B. H., et al. (2012). Adult outcomes as a function of an early childhood educational program: An Abecedarian Project follow-up. *Developmental Psychology*, 48, 1033–1043.
- Clements, D. H., & Sarama, J. (2007). Effects of a preschool mathematics curriculum: Summative research on the Building Blocks project. *Journal for Research in Mathematics Education*, 38, 136–163.
- Clements, D. H., & Sarama, J. (2011). Early childhood mathematics intervention. *Science*, 333(6045), 968–970.
- Clements, D. H., Sarama, J., & Germeroth, C. (2016). Learning executive function and early mathematics: Directions of causal relations. *Early Childhood Research Quarterly*, 36, 79–90.
- Clements, D. H., Sarama, J., Spitler, M. E., Lange, A. A., & Wolfe, C. B. (2011). Mathematics learned by young children in an intervention based on learning trajectories: A large-scale cluster randomized trial. *Journal for Research in Mathematics Education*, 42, 127–166.
- Clements, D. H., Sarama, J., Wolfe, C. B., & Spitler, M. E. (2013). Longitudinal evaluation of a scale-up model for teaching mathematics with trajectories and technologies: Persistence of effects in the third year. *American Educational Research Journal*, 50, 812–850.
- Copley, J. (2004). The early childhood collaborative: A professional development model to communicate and implement the standards. In D. Clements, J. Sarama, & A. M. DiBiase (Eds.), *Engaging young children in mathematics: Standards for early childhood mathematics education* (pp. 401–414). Mahwah, NJ: Erlbaum.
- Copple, C., & Bredekamp, S. (2009). *Developmentally appropriate practice in early childhood programs serving children from birth to age eight*. Washington, DC: NAEYC.
- Denton, K., & West, J. (2002). *Children's reading and mathematics achievement in kindergarten and first grade*. Washington, DC: National Center for Education Statistics.
- Derman-Sparks, L., & Moore, E. (2016). Two teachers look back: The Ypsilanti Perry Preschool. Part I. *Young Children*, 71, 4.
- Dewey, J. (1938/1963). *Experience and education*. New York: Collier Books.
- Dodge, K. A., Bai, Y., Ladd, H. F., & Muschkin, C. G. (2017). Impact of North Carolina's early childhood programs and policies on educational outcomes in elementary school. *Child Development*, 88, 996–1014.
- Domitrovich, C. E., Cortes, R. C., & Greenberg, M. T. (2007). Improving young children's social and emotional competence: A randomized trial of the Preschool "PATHS" Curriculum. *The Journal of Primary Prevention*, 28, 67–91.
- Domitrovich, C. E., Gest, S. D., Gill, S., Bierman, K. L., Welsh, J. A., & Jones, D. (2009). Fostering high-quality teaching with an enriched curriculum and professional development support: The Head Start REDI program. *American Educational Research Journal*, 46(2), 567–597.
- Duncan, G. J., Dowsett, C. J., Claessens, A., Magnuson, K., Huston, A. C., Klebanov, P., et al. (2007). School readiness and later achievement. *Developmental Psychology*, 43(6), 1428–1446.
- Epstein, A. E. (2014). *Essentials of active learning in preschool* (2nd ed.). Ypsilanti, MI: HighScope Press.
- Epstein, A. E. (2007). *Numbers Plus preschool mathematics curriculum*. Ypsilanti, MI: HighScope Press.
- Frede, E., Jung, K., Barnett, W. S., & Figueras, A. (2009). *The APPLES Blossom: Abbott preschool program longitudinal effects study (APPLES) preliminary results through 2nd grade*. New Brunswick, NJ: National Institute for Early Education Research.
- Frede, E. C., Jung, K., Barnett, W. S., Lamy, C. E., & Figueras, A. (2007). *The Abbott preschool program longitudinal effects study. Report to the New Jersey Department of Education*. New Brunswick, NJ: NIEER.
- Fuson, K. C., Clements, D. H., & Beckmann, S. (2010). *Focus in prekindergarten: Teaching with curriculum focal points*. Reston, VA: National Council for Teachers of Mathematics, Washington, D.C.: National Association for the Education of Young Children.
- Gelman, R., & Brenneman, K. (2004). Science learning pathways for young children. *Early Childhood Research Quarterly*, 19, 10–18.
- Gelman, A., & Hill, J. (2007). *Data analysis using regression and multilevel/hierarchical models*. Cambridge, U.K: Cambridge University Press.
- Gerde, H. K., Pierce, S. J., Lee, K., & Van Egeren, L. A. (2018). Early childhood educators' self-efficacy in science, math, and literacy instruction and science practice in the classroom. *Early Education and Development*, 29(1), 70–90.
- Gilliam, W., & Zigler, E. (2004). *State efforts to evaluate the effects of prekindergarten: 1977 to 2003*. Unpublished manuscript New Haven Yale University Child Study Center.
- Ginsburg, H. P., Kaplan, R. G., Cannon, J., Cordero, M. I., Eisenband, J. G., Galanter, M., et al. (2006). Helping early childhood educators to teach mathematics. In M. Zaslow, & I. Martinez-Beck (Eds.), *Critical issues in early childhood professional development* (pp. 171–202). Baltimore, MD: Paul H. Brookes.
- Ginsburg, H. P., & Pappas, S. (2016). Invitation to the birthday party: Rationale and description. *ZDM Mathematics Education*, 48(7), 947–960.
- Ginsburg, H., Pappas, S., Lee, Y. S., & Chiong, C. (2010). *Early Mathematics Assessment System (EMAS)*. Teacher's College, Columbia University.
- Gormley, W. T., Gayer, T., Phillips, D., & Dawson, B. (2005). The effects of universal pre-k on cognitive development. *Developmental Psychology*, 41(6), 872–884.
- Gormley, W. T., Phillips, D., & Gayer, T. (2008). Preschool programs can boost school readiness. *Science*, 320, 1723–1724.
- Griffin, J. A. (2010). Research on the implementation of preschool intervention programs: Learning by doing. *Early Childhood Research Quarterly*, 25, 267–269. <http://dx.doi.org/10.1016/j.ecresq.2010.03.004>
- Halle, T. G., & Darling-Churchill, K. E. (2016). Review of measures of social and emotional development. *Journal of Applied Developmental Psychology*, 45, 8–18.
- Heckman, J. J., & Karakapula, G. (2019). *Intergenerational and intragenerational externalities of the Perry Preschool Project*. Working Paper 2019-033. https://econresearch.uchicago.edu/sites/econresearch.uchicago.edu/files/Heckman_Karakapula.2019.inter-intra-externalities-perry-r2.pdf
- Hegland, S., & Oesterreich, L. (2011). June Individualizing coaching: Matching coaching strategies to teacher understanding and motivation. In *Paper presented at the National Association for the Education of Young Children (NAEYC) Professional Development Institute*.
- HighScope Educational Research Foundation. (2014). *COR advantage 1.5*. Ypsilanti, MI: The HighScope Press.
- HighScope Educational Research Foundation. (2003a). *Preschool program quality assessment*. Ypsilanti, MI: The HighScope Press.
- HighScope Educational Research Foundation. (2003b). *Preschool program quality assessment: Administrator's manual*. Ypsilanti, MI: The HighScope Press.
- Hresko, W., Peak, P., Herron, S., & Bridges, D. (2000). *Young children's achievement test*. Austin, TX: Pro-Ed, Incorporated.
- Institute of Education Sciences. (2017). *What Works Clearinghouse standards handbook version 4.0*. https://ies.ed.gov/ncee/wwc/Docs/referenceresources/wwc_standards_handbook.v4.pdf
- Jenkin, J. M., & Duncan, G. J. (2017). Do pre-kindergarten curricula matter? The current state of scientific knowledge on pre-kindergarten effects, 37–44. https://www.brookings.edu/wp-content/uploads/2017/04/duke_prekstudy_final_4-4-17_hires.pdf
- Jeynes, W. H. (2006). Standardized tests and Froebel's original Kindergarten model. *Teachers College Record*, 108(10), 1937–1959.
- Kaufman, A. S., & Kaufman, N. L. (1985). *Kaufman test of educational achievement*. Circle Pines, MN: American Guidance Service.
- Klein, A., Starkey, P., Clements, D., Sarama, J., & Iyer, R. (2008). Effects of a pre-kindergarten mathematics intervention: A randomized experiment. *Journal of Research on Educational Effectiveness*, 1, 155–178.
- Ladd, H. F., Muschkin, C. G., & Dodge, K. A. (2014). From birth to school: Early childhood initiatives and third-grade outcomes in North Carolina. *Journal of Policy Analysis and Management*, 33, 162–187.
- Lee, J. S., & Ginsburg, H. P. (2007). What is appropriate mathematics education for four-year-olds? Pre-kindergarten teachers' beliefs. *Journal of Early Childhood Research*, 5(1), 2–31.
- Lee, Y. S. (2016). Psychometric analyses of the Birthday Party. *ZDM Mathematics Education*, 48(7), 961–975.
- Lipsey, M. W., Farran, D. C., & Hofer, K. G. (2016). *Effects of a state prekindergarten program on children's achievement and behavior through third grade working paper*. Peabody Research Institute. <https://peabody.vanderbilt.edu/research/pri/TNVPK.Grade.3.working-paper.pdf>
- Love, J. M., Chazan-Cohen, R., & Raikes, H. (2007). Forty years of research knowledge and use: From Head Start to Early Head Start and beyond. In J. L. Aber, S. J. Bishop-Josef, S. M. Jones, K. T. McLearn, & D. A. Phillips (Eds.), *Child development and social policy: Knowledge for action* (pp. 79–95). Washington, DC: American Psychological Association.
- Mackintosh, B. B., & McCoy, D. C. (2019). Exploring social competence as a mediator of head start's impact on children's early math skills: Evidence from

- the Head Start Impact Study. *Early Education and Development*. Advance online publication. <https://doi.org/10.1080/10409289.2019.1576156>.
- McGrew, K. S., & Woodcock, R. W. (2001). *Technical manual*. Woodcock-Johnson III. Itasca, IL: Riverside Publishing.
- National Center on Quality Teaching and Learning. (2015). *Preschool curriculum consumer report*. <https://eclkc.ohs.acf.hhs.gov/sites/default/files/pdf/curriculum-consumer-report.pdf>
- National Council of Teachers of Mathematics. (2006). *Curriculum focal points for prekindergarten through grade 8 mathematics: A quest for coherence*. Reston, VA: Author.
- National Association for the Education of Young Children/National Council of Teachers of Mathematics (NAEYC/NCTM). (2010). *Early childhood mathematics: Promoting good beginnings*. <https://www.naeyc.org/sites/default/files/globally-shared/downloads/PDFs/resources/position-statements/psmath.pdf>
- National Research Council. (2005). *How students learn science in the classroom*. Washington, DC: National Academies Press.
- National Research Council. (2009). *Mathematics learning in early childhood: Paths toward excellence and equity*. Washington, DC: National Academies Press.
- Natriello, G., McDill, E. L., & Pallas, A. M. (1990). *Schooling disadvantaged children: Racing against catastrophe*. New York: Teachers College Press.
- Neill, P., & Gainsley, S. (2014). *Meaningful math in preschool: Making math count throughout the day*. Ypsilanti, MI: HighScope Press.
- Parks, A. N., Wakabayashi, T., & Hardin, B. A. (2018). Highlighting meaningful counting in prekindergarten. *Teaching Children Mathematics*, 25(1), 53–59.
- Petersen, E. (1893). Educating the whole child. *The Phrenological Journal and Science of Health*, 96, 27.
- Piaget, J., & Inhelder, B. (1969). *The psychology of the child*. New York: Basic Books.
- Presser, A. L., Clements, M., Ginsburg, H., & Ertle, B. (2015). Big Math for Little Kids: The effectiveness of a preschool and kindergarten mathematics curriculum. *Early Education and Development*, 26, 399–426.
- Puma, M., Bell, S., Cook, R., Heid, C., Broene, P., Jenkins, F., et al. (2012). *Third grade follow-up to the head start impact study: Final report*. Washington, DC: Office of Planning, Research and Evaluation, Administration for Children and Families. https://www.acf.hhs.gov/sites/default/files/opre/head_start_report.pdf
- Ramey, C. T. (2018a). The abecedarian approach to social, educational, and health disparities. *Clinical Child and Family Psychology Review*, 21(4), 527–544.
- Ramey, C. T. (2018b). Correction to: The abecedarian approach to social, educational, and health disparities. *Clinical Child and Family Psychology Review*, 21(4), 545.
- Raudenbush, S. W., & Bryk, A. S. (2002). *Hierarchical linear models: Applications and data analysis methods* (2nd ed.). Thousand Oaks, CA: Sage Publications, Inc.
- Raver, C. (2002). Emotions matter: Making the case for the role of young children's emotional development for early school readiness. *Social Policy Report of the Society for Research in Child Development*, 16, 1–20.
- Raver, C. C., & Zigler, E. F. (1997). Social competence: An untapped dimension in evaluating Head Start's success. *Early Childhood Research Quarterly*, 12, 365–385.
- Reyna, V. F., Nelson, W. L., Han, P. K., & Dieckmann, N. F. (2009). How numeracy influences risk comprehension and medical decision making. *Psychological Bulletin*, 135, 943–973.
- Reynolds, A. J., & Temple, J. A. (1998). Extended early childhood intervention and school achievement: Age 13 findings from the Chicago Longitudinal Study. *Child Development*, 69, 231–246.
- Reynolds, A. J., Temple, J. A., & Ou, S. (2010). Preschool education, educational attainment, and crime prevention: Contributions of cognitive and non-cognitive skills. *Children and Youth Services Review*, 32, 1054–1063.
- Reynolds, A. J., Temple, J. A., White, B. A., Ou, B., & Robertson, S. D. L. (2011). Age 26 cost-benefit analysis of the Child-Parent Center early education program. *Child Development*, 82, 379–404.
- Ritchie, S. J., & Bates, T. C. (2013). Enduring links from childhood mathematics and reading achievement to adult socioeconomic status. *Psychological Science*, 24, 1301–1308.
- Sanford DeRousie, R. M., & Bierman, K. L. (2012). Examining the sustainability of an evidence-based preschool curriculum: The REDI program. *Early Childhood Research Quarterly*, 27, 55–65.
- Sarama, J., Clements, D. H., Wolfe, C. B., & Spitler, M. (2012). Longitudinal evaluation of a scale-up model for teaching mathematics with trajectories and technologies. *Journal of Research on Educational Effectiveness*, 5, 105–135.
- Schweinhart, L. J., Montie, J., Xiang, Z., Barnett, W. S., Belfield, C. R., & Nores, M. (2005). *Lifetime effects: The HighScope Perry Preschool Study through age 40*. Ypsilanti, MI: The HighScope Press.
- Schweinhart, L. J., Xiang, Z., Daniel-Echols, M., Browning, K., & Wakabayashi, W. (2012). *Michigan Great Start Readiness Program evaluation 2012: High school graduation and grade retention findings*. Ypsilanti, MI: HighScope Educational Research Foundation.
- Shore, R. (1997). *Rethinking the brain*. Washington, DC: Families and Work Institute.
- Shonkoff, J. P., & Phillips, D. A. (2000). *From neurons to neighborhoods: The science of early childhood development*. Washington, DC: National Academy Press.
- Snijders, T. A. B., & Bosker, R. J. (2012). *Multilevel analysis: An introduction to basic and advanced multilevel modeling* (2nd ed.). London, UK: Sage Publishers.
- Spybrook, J. (2012). *June Examining alternative design schemes for the Numbers Plus Efficacy Study*. Unpublished design memo.
- Spybrook, J., & Raudenbush, S. W. (2009). An examination of the precision and technical accuracy of the first wave of group-randomized trials funded by the institute of education sciences. *Educational Evaluation and Policy Analysis*, 31(3), 298–318.
- Starkey, P., Klein, A., & Wakely, A. (2004). Enhancing young children's mathematical knowledge through a pre-kindergarten mathematics intervention. *Early Childhood Research Quarterly*, 19, 99–120.
- Thompson, R. A., & Nelson, C. A. (2001). Developmental science and the media: Early brain development. *American Psychologist*, 56, 5–15.
- U.S. Department of Education. (2007). *SRA Real Math Building Blocks PreK. What Works Clearinghouse intervention report*. https://ies.ed.gov/ncee/wwc/Docs/InterventionReports/WWC_Building.Blocks.072307.pdf
- U.S. Department of Health and Human Services. (2016). *Head Start program performance standards*. <https://eclkc.ohs.acf.hhs.gov/sites/default/files/pdf/hspss-appendix.pdf>
- Vygotsky, L. S. (1934/1962). *Language and thought*. Cambridge, MA: M.I.T. Press.
- Wakabayashi, T., Claxton, J., & Smith, E. V. (2017). Validation of a revised observation-based assessment tool for children birth through kindergarten: The COR Advantage. *Journal of Psychoeducational Assessment*, <http://dx.doi.org/10.1177/0734282917732491>
- Wechsler, D. (1992). *Wechsler individual achievement test*. San Antonio, TX: Psychological Corporation.
- Weikart, D. P., Rogers, L., Adcock, C., & McClelland, D. (1970). *The cognitively oriented curriculum: A framework for preschool teachers*. Ypsilanti, MI: HighScope Press.
- Weiland, C., McCormick, M., Mattern, S., Maier, M., & Morris, P. (2018). Preschool curricula and professional development features for getting to high-quality implementation at scale: A comparative review across five trials. *AERA Open*, 4(1), 1–16.
- Weiland, C., & Yoshikawa, H. (2013). Impacts of a prekindergarten program on children's mathematics, language, literacy, executive function, and emotional skills. *Child Development*, 84, 2112–2130.
- Woodcock, R. W., McGrew, K. S., & Mather, N. (2001). *Woodcock-Johnson III Tests of Achievement* (3rd ed.). Itasca, IL: Riverside.
- Zigler, E. F., & Trickett, P. K. (1978). IQ, social competence, and evaluation of early childhood intervention programs. *American Psychologist*, 33, 789–798.